

Team Details

a. Team name: Hey Merlion!
b. Team leader name: Changhyeon Nam (team: Changhyeon Nam, Hyunho Yang)
c. Problem Statement: BFSI: Intelligent Risk, Fraud & Financial Experiences

Why this matters

THE PROBLEM

Banks are ready to lend on cash flow. But manual review costs too much, and numbers from a basic AI prompt cannot be traced to a page.

MEET LINH

Shopee seller, Vietnam

Needs VND 150M in working capital. Her sales are real.

She brings 14 documents: a bank statement PDF, monthly Shopee dashboard screenshots, supplier receipts, photos of the shop.

No credit file. No collateral.
Today she is declined, or priced blindly.

Synthetic demo case B

1. **8.8%** of Vietnam's SMEs have a bank loan; large firms: 47.1%. Without a credit file or collateral, small firms are declined.
2. **75.5%** of Vietnamese firms cannot borrow without collateral, so banks are moving to cash-flow lending (VPBank, Vietcombank, Agribank).
3. **~45 min** to read and verify one file by hand. For a small loan, the reading costs more than the loan earns.
4. **77%** of memo numbers from a single Gemini prompt have no source page. The underwriter must find each one again. That prompt also caught only 15% of the planted risks.

Sources: 1 FiinGroup via Viet Nam News (Aug 2026) · 2 VCCI 2025 via VTV (May 2026) · 3 team estimate; ADB Asia SME Monitor 2025 · 4 our live benchmark, gemini-3.8-flash

The solution

THE SOLUTION

LedgerLens reads the records, shows its work and challenges its findings before a human signs.

1 Read with evidence

CUTS READING COST

Gemini reads every page. Each value has a document, page and bounding box. Python computes the cash flow. A memo with an unsourced number is rejected and regenerated.

In Linh's file: 14 documents, every figure sourced to a page and box, in about 2 minutes.

2 Challenge

CHECKS AI OUTPUT

A separate agent never sees the writer's prompt. It runs deterministic checks and 12 underwriting questions, then ranks objections with evidence.

In Linh's file: Flags a reused receipt photo and VND 82.5M of own-account transfers counted as income.

3 Human signs

KEEPS ACCOUNTABILITY

The underwriter accepts or dismisses each objection and signs. Every agent call is logged; decisions set priorities for the next run.

In Linh's file: Linh receives the decision and its conditions in Vietnamese.

Grounded only in the applicant's documents. Guardrails: output schema, deterministic checks, a second agent, a human signature.

Opportunities

EXPECTED IMPACT

What changes for each team.

For the lender

45 → 2

minutes per file

USD ~11 → 0.28 per case. Small loans become worth underwriting.

For the small business

Cash flow

not collateral, decides the loan

Assessed using her existing records. A decline comes with reasons in her own language.

For risk and audit

77% → 0%

of memo numbers with no page

Every figure links to a page and box, so nothing needs manual re-checking. Planted risks caught: 90% vs 15%.

USP Separate agents write and verify · code computes numbers, never generates them · every number links to its source line · decisions set check priorities

Back to the 8.8%: when reading a file costs cents and minutes instead of an hour, the loans banks decline today become worth making.

The manual baseline (45 min at USD 15/h) is a team estimate. A number is untraceable if it has no source document and page; correctness was not measured. Live gemini-3.8-flash on a small synthetic sample.

Features

Every feature is part of one pipeline that ends with a human decision.

Intake → Extract → Cash flow → Analyze → **Challenge** → Decide → Explain

Multimodal intake

PDF, screenshots, photos. Classified, with quality flags.

Evidence-linked extraction

Doc id, page, bbox on every value. PDF text layer verifies it.

Deterministic model

Inflow, outflow, trend, volatility, debt service. Python, not the model.

Clickable memo

Cites only catalog figures. Click to open the source line.

Tools + 12 checks

Image hash, balance chain, dates. Ranked objections with evidence.

Underwriter workbench

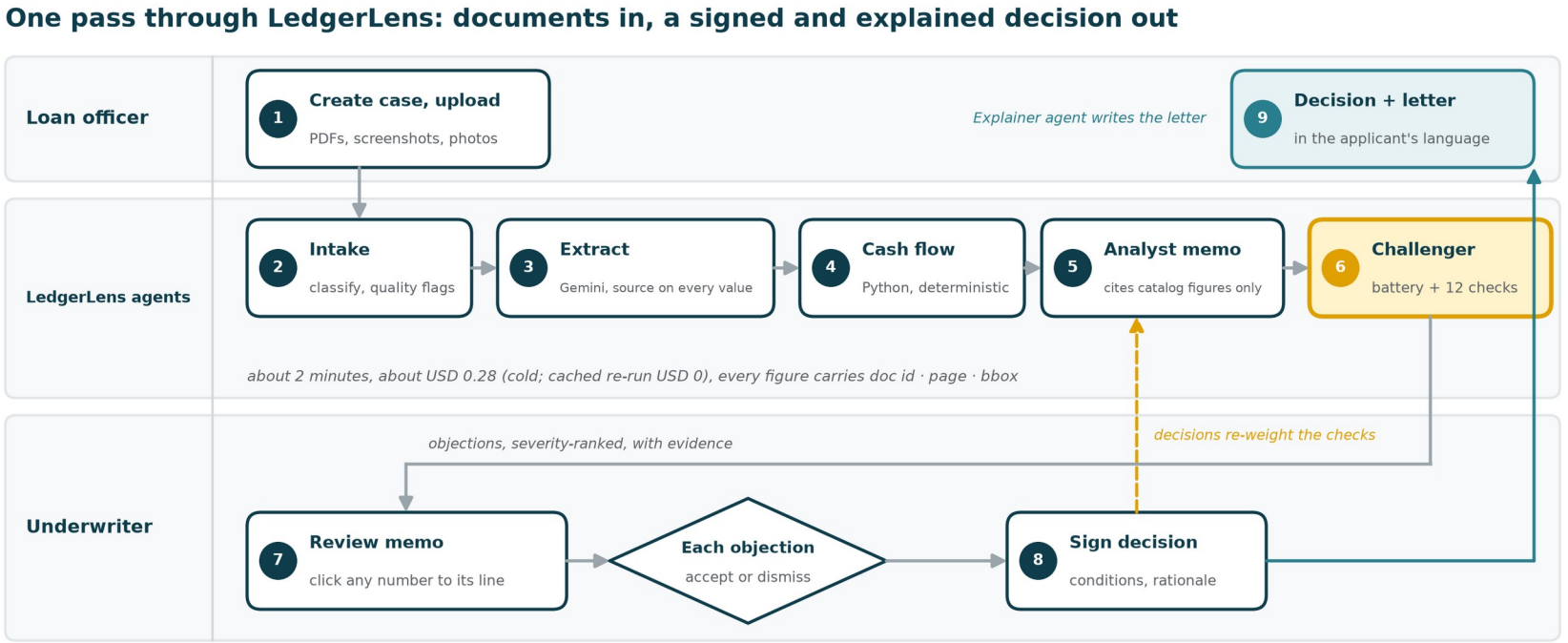
Accept or dismiss, then sign. Audit log. Decisions guide checks.

Letter and print memo

Customer letter in the local language. Memo prints with footnoted sources.

Out of scope by design: credit scoring, loan pricing, core-banking integration. LedgerLens informs the decision; it does not make it.

Process flow diagram

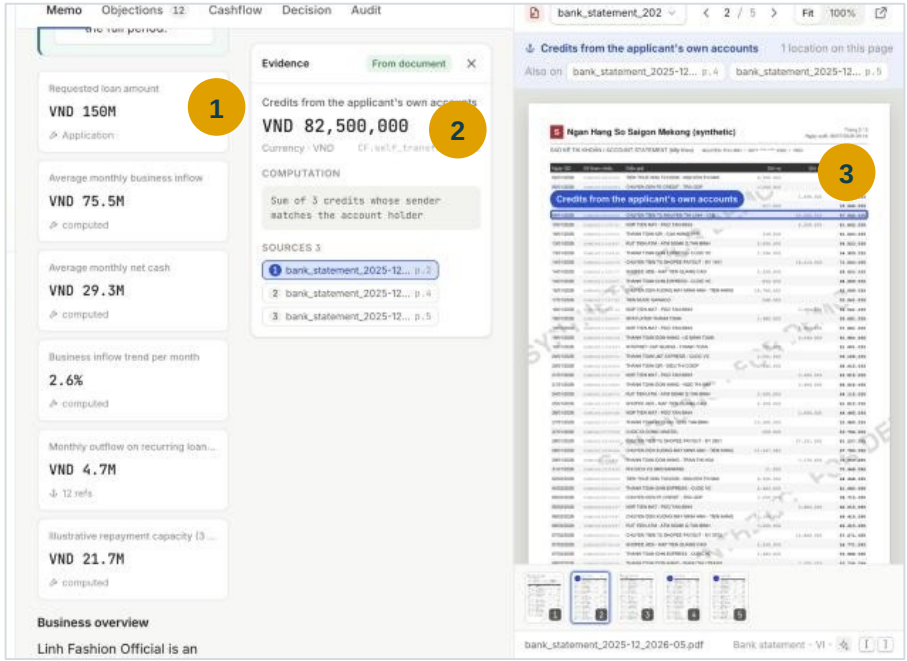


Three lanes: the loan officer uploads, the agents read and challenge, the underwriter decides. Decisions flow back into the Challenger's priorities.

The product in use

THE PRODUCT IN ONE INTERACTION

Click the number. See the line.



1 The memo cites a figure

Every underlined value is a catalog figure with a source. Here: VND 82,500,000, credits from the applicant's own accounts.

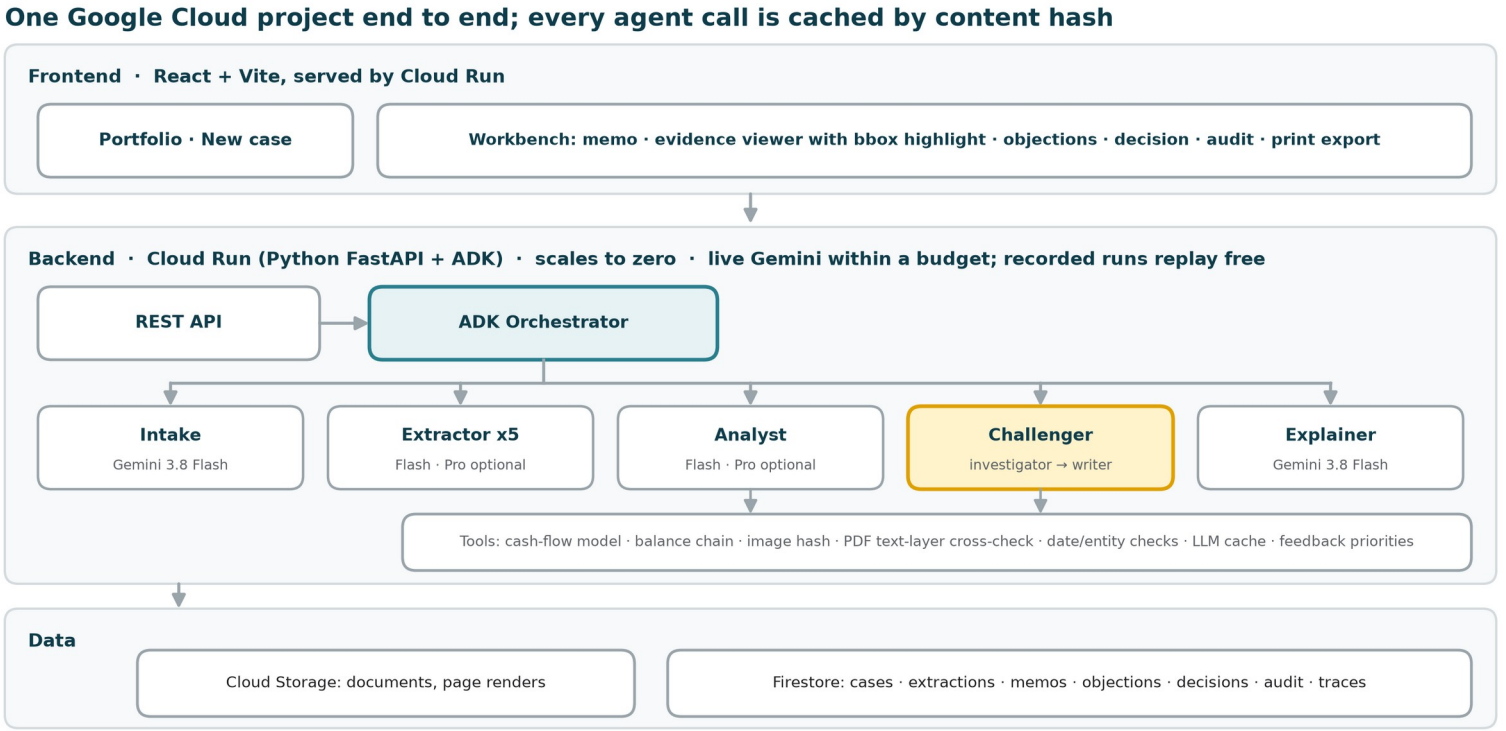
2 One click opens the evidence

The calculation: sum of 3 credits whose sender matches the account holder. All 3 sources: pages 2, 4 and 5.

3 The source line lights up

The statement page opens with the extractor's bounding box drawn on the exact row. A memo with an unsourced number is rejected and regenerated.

Architecture



Technology

Gemini reads, ADK orchestrates, Python computes, Cloud Run serves.

GOOGLE CLOUD

Gemini 3.8 Flash

Default for every agent. Measured USD 0.26-0.31 per cold case; USD 0 cached.

GOOGLE CLOUD

Gemini 3.1 Pro

One env var switches Extractor, Analyst and Challenger to Pro.

GOOGLE CLOUD

ADK on Cloud Run

Agent Platform: Intake, 5 Extractors, Analyst, Challenger (investigator, writer), Explainer.

GOOGLE CLOUD

Cloud Run

One service: FastAPI, the ADK agents and the React workbench; scales to zero.

GOOGLE CLOUD

Firestore + Storage

Cases, extractions, memos, objections, decisions, audit, traces; documents.

PYTHON

Deterministic tools

Cash-flow model, balance chain, image hash, PDF text-layer cross-check, date and entity checks.

PYTHON

LLM cache, budget cap

Content-hash cache per model and prompt version; USD cap shared by all instances.

INTEGRATION

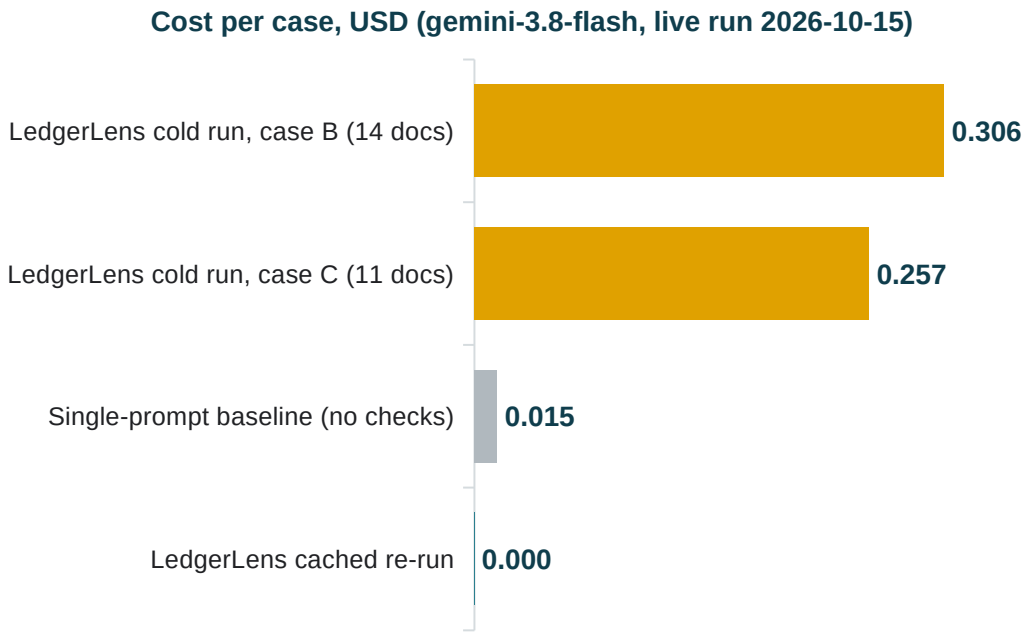
REST API

Any loan origination system can create cases, run stages and read memos.

Grounding: only the applicant's documents; every value must carry source {doc_id, page, bbox}. No separate OCR: Gemini reads rendered pages.

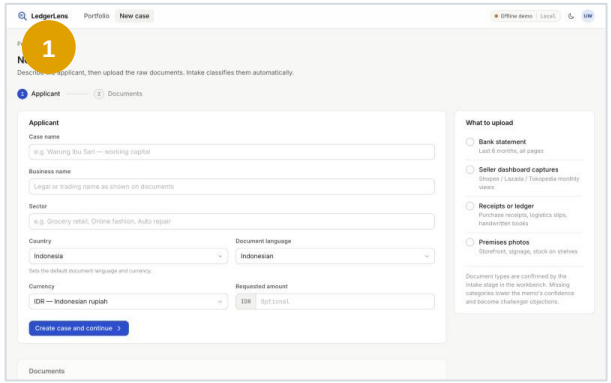
Estimated implementation cost

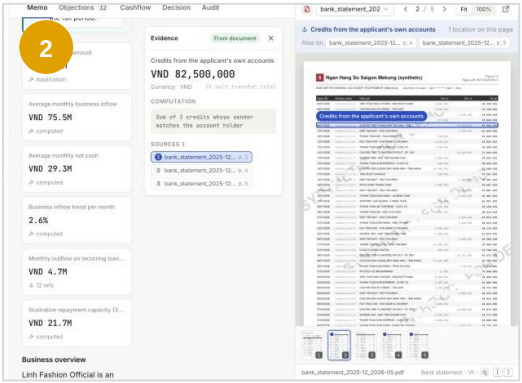
Measured: USD 0.26-0.31 per case on a cold Flash run, USD 0 on a cached re-run

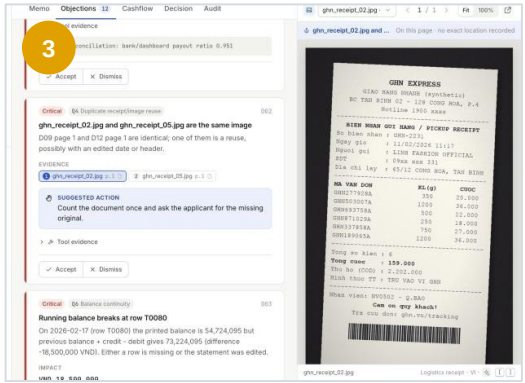


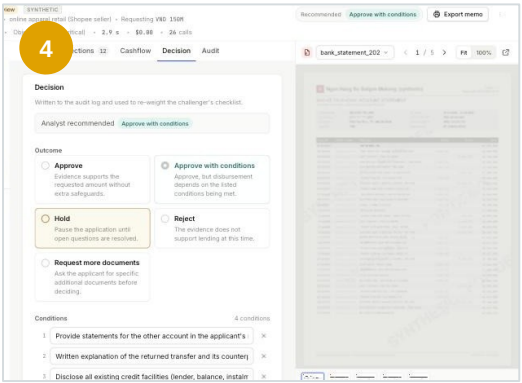
Assumption	Value
Gemini 3.8 Flash list price	USD 0.75 / 3.75 per 1M tokens
Measured cold run	~2 min and USD 0.26-0.31 per case
Cached re-run	USD 0 (content-hash LLM cache)
Public demo	Replays the recorded live Gemini run, USD 0; only the customer letter calls Gemini (~USD 0.0015)
Hosting	Cloud Run scales to zero; Firestore and Storage in free tier
Guardrail	Budget-cap env var stops Gemini calls at the limit
Build	2 engineers, 4 days, free tier and trial credits

Prototype screenshots









1 New case

Applicant details, then raw documents. Intake classifies them and flags quality issues.

2 Click a number, see the line

Evidence panel with computation and all sources; the statement opens with the bounding box highlighted.

3 Challenger objections

Critical: two receipt photos are the same image with an edited date. Evidence chips open the photo.

4 The underwriter signs

Outcome, conditions and rationale go into the audit log; accept/dismiss rates set check priorities.

Prototype benchmark

LedgerLens caught 90% of the planted risks on live Gemini output; a single prompt on the same documents caught 15%.

LEDGERLENS

90%

of planted risks caught

small synthetic sample; indicative results

SINGLE PROMPT

15%

same documents

and same model

100%

of memo numbers link to a page (single prompt: 23%)

~2 min

per cold case, USD 0.26-0.31; cached re-run USD 0

Metric	Target	Result
Risks caught (strict)	≥ 80%	90% (single prompt: 15%)
Traceable numbers	100%	100% (single prompt: 23%)
Objections on clean case A	≤ 20%	none
Extraction row accuracy*	≥ 97%	100%
Evidence-link accuracy*	≥ 95%	100%
Time per case (cold)	≤ 180 s	~2 min
Cost per case (cold)	≤ USD 0.50	USD 0.26-0.31

Measurement. Live gemini-3.8-flash runs, 2026-10-15/16, on three synthetic businesses with team-planted risks. Strict scoring: each objection must include the risk's identifying amount or document, one objection per risk. Traceable = a memo number linked to a document and page; correctness not measured. The sample is small, so every percentage is indicative. Missed: dashboard gross vs payout, COD vs cash deposits. Recommendation matched on C; A gave approve_with_conditions (expected approve), B request_more_documents (expected approve_with_conditions). *Clean born-digital synthetic PDFs; real scans less accurate.

Future development

Where this goes next, and what we do not claim yet

Pilot Q1 2027

One SEA bank or MFI, real consented SME files; underwriter decisions replace synthetic traps

→

Integrate Q2–Q3 2027

Open-banking feeds (Brankas, Finverse) beside PDFs; hooks into loan origination systems

→

Scale 2028

Same pattern for trade finance and micro-insurance; shared Challenger check library

What we do not claim yet, and which phase answers it

Synthetic benchmark

A small synthetic sample with planted risks; real-data results come with the pilot.

Forged documents

Hash and balance-chain checks catch crude edits. Open-banking data closes the gap for the rest.

Platform sellers

Shopee and Grab already lend to their own sellers. We target banks and MFIs lending outside platforms.

positioning, not a feature

Humans decide. Every figure and decision is traceable. The prototype uses only watermarked synthetic data. Production keeps documents and model endpoints in-region.

Submission links

GitHub public repository

github.com/changhyeonnam/ghack

English README, architecture, run instructions, synthetic-data notice, presentation/

Demo video (2 min 59 s)

TBD

Problem, how it works, one case end to end (click-to-source, Challenger, decision, letter), benchmark

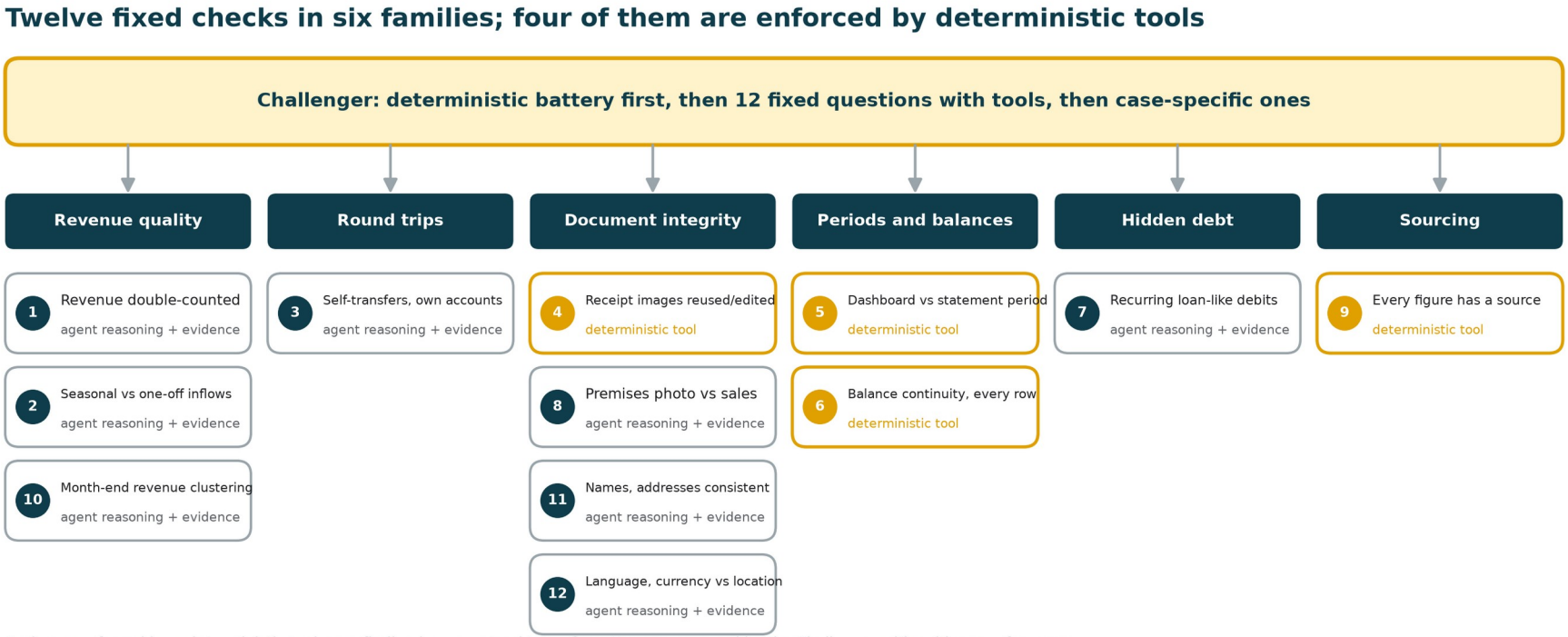
Final product

ledgerlens-api-629502893425.asia-southeast1.run.app

Cloud Run, live Gemini within a fixed AI budget (on-screen notice when reached). Each browser gets its own copy of the three demo cases, which replay the recorded live run in seconds; visitors' cases are shared read-only. No login

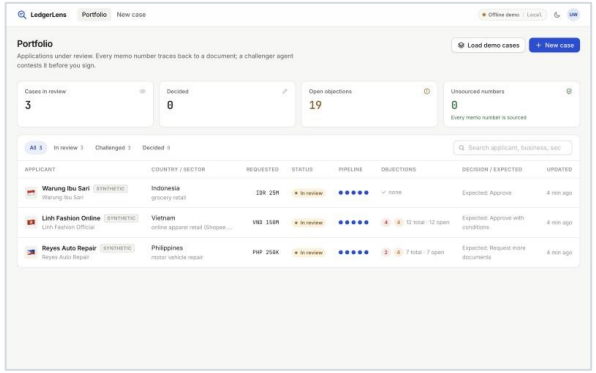
Links are inserted at submission on 18 October 2026.

Appendix: the Challenger's 12 fixed checks



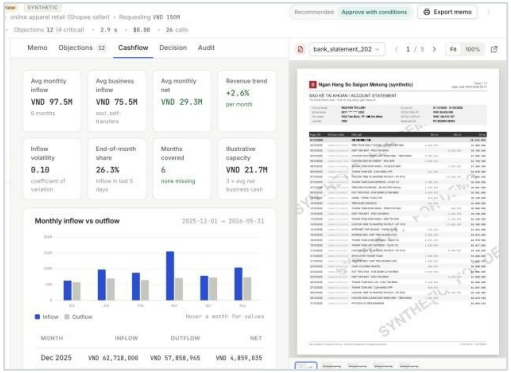
Amber = enforced by a deterministic tool, so a finding is guaranteed to surface. Grey = reasoned by the Challenger with evidence references.

Appendix: portfolio, cash flow and audit views



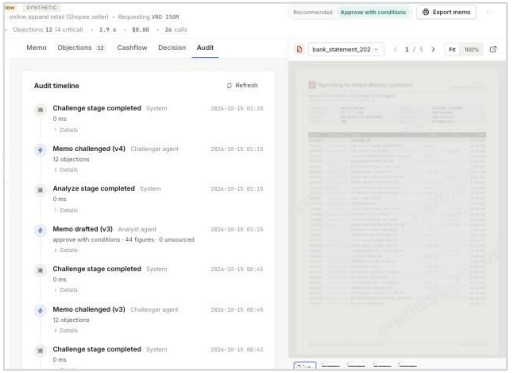
Portfolio

Cases in review, open objections and unsourced numbers across the book; filters by status.



Cash flow

Deterministic monthly model: inflow, net, trend, volatility, end-of-month share, repayment capacity.



Audit timeline

Every stage and agent call, with version, model and prompt, in a trail regulators can review.

Also in the build: dark and light themes, PDF text-layer cross-check on every extracted amount, content-hash LLM cache, cache_only demo mode, budget cap, printable paper memo with footnoted sources.

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

